

CHAPTER 1

INTRODUCTION

Background of the Study

The popularity of mobile games has exponentially increased in today's era, to the point that traditional methods of teaching history can be transformed into game-based formats, making history more engaging and interesting (Zirawaga et al., 2017). History-based games have emerged as an alternative approach to enhance learner satisfaction and address the boredom of repetitive memorization. They offer an engaging and immersive learning experience that is more appealing than conventional instructional methods. According to Fahrudin and Saefudin (2025), the use of interactive media and primary sources in history learning increases student participation and retention by transforming passive learning into an engaging experience.

In the Philippines, history-based game applications have proven effective in enhancing student engagement and participation. For example, *HiStorya*, a game-based mobile learning application for Grade 8 Araling Panlipunan students, utilized mobile technology to motivate learners to study Philippine history both inside and outside the classroom. Results showed that the game encouraged active participation and improved students' interest in historical topics (Miguel et al., 2015). Similarly, *KaSAYSAYan: The Integration of Philippine History as an Interactive and Informative Mobile Game Application* (2023), built using the Unity Game Engine, was designed to make



learning Philippine history more engaging and interactive. Findings revealed that the game increased students' motivation and made learning more relevant and enjoyable for elementary students studying Philippine history (Centeno et al., 2023). These studies demonstrate that history-based games serve as innovative tools for transforming history education from a passive, memorization-driven subject into an active, participatory, and meaningful learning experience.

The idea behind the development of *Chronoquest: The Pangasinan Chronicles* was driven by the developers' observation that many students and young learners find history boring and uninteresting due to heavy reliance on rote memorization, which often leads to disengagement. The existing curriculum in teaching history still depends heavily on textbook-based instruction and memorization. This traditional approach limits students' critical thinking and results in poor classroom participation, as learners tend to focus only on memorizing facts, dates, and events instead of engaging in meaningful discussions. Numerous studies have confirmed the effectiveness of history-themed games, which have long been popular among players due to their rich narratives and immersive storytelling. Unlike other types of games, historical games use real-world events as their backdrop, which not only supports the storyline but also enriches educational content.

By developing *Chronoquest: The Pangasinan Chronicles*, the researchers aim to provide learners with a different experience of exploration through Western Pangasinan history, addressing challenges faced by many Filipinos in studying history. The game will feature interactive gameplay, 3D



models, and educational quests and missions, allowing users to appreciate their heritage, connect with their community, and contribute to the preservation of the rich legacy of Western Pangasinan. By making history come alive through technology, *Chronoquest: The Pangasinan Chronicles* seeks to promote a deeper appreciation for Western Pangasinan's heritage and inspire a new generation of developers.

Conceptual Framework

This study is based on the Input–Process–Output (IPO) model, which serves as both the research paradigm and the conceptual framework. The IPO model illustrates how the proposed game, *Chronoquest: The Pangasinan Chronicles*, is designed, developed, and evaluated.

Input. Data were collected through a survey conducted among students who find history uninteresting, to gather insights about their interest and engagement in learning local history. In addition, historical information about Western Pangasinan was obtained from the internet, ensuring credibility by consulting legitimate and authentic sources. These inputs guided the design of *Chronoquest: The Pangasinan Chronicles* to make history learning more engaging and meaningful.

Process. The Agile Software Development Life Cycle (SDLC) was applied in the development of the game. The process began with the **Concept phase**, where the developers brainstormed and collaborated on ideas to promote Western Pangasinan's history through an interactive Android-based game that



would address the identified problem. During the **Inception phase**, the developers identified and planned the game's objectives, features, storyline, and interface design based on the results of the surveys conducted among the target user's preferences. In the **Iteration phase**, the developers designed and created a playable version featuring Bolinao as the first iteration, demonstrating the game's core functionality. In the following sprints or iterations, additional municipalities were designed and added. After each addition of municipalities, the developers test the game to ensure that the game was functioning properly and free of errors. The iteration process continued until all featured municipalities were test, review and playable and ready for release. Finally, in the Release phase, the developers produced a functional version of Chronoquest: The Pangasinan Chronicles.

Output. A playable version of Chronoquest: The Pangasinan Chronicles was developed. The game allows players to explore the two municipalities of Western Pangasinan and create an interactive experience through storytelling, missions, and quests, encouraging users to appreciate their local history and ancestors.

Figure 1 shown on the next page presents the conceptual framework of the study using the IPO model.

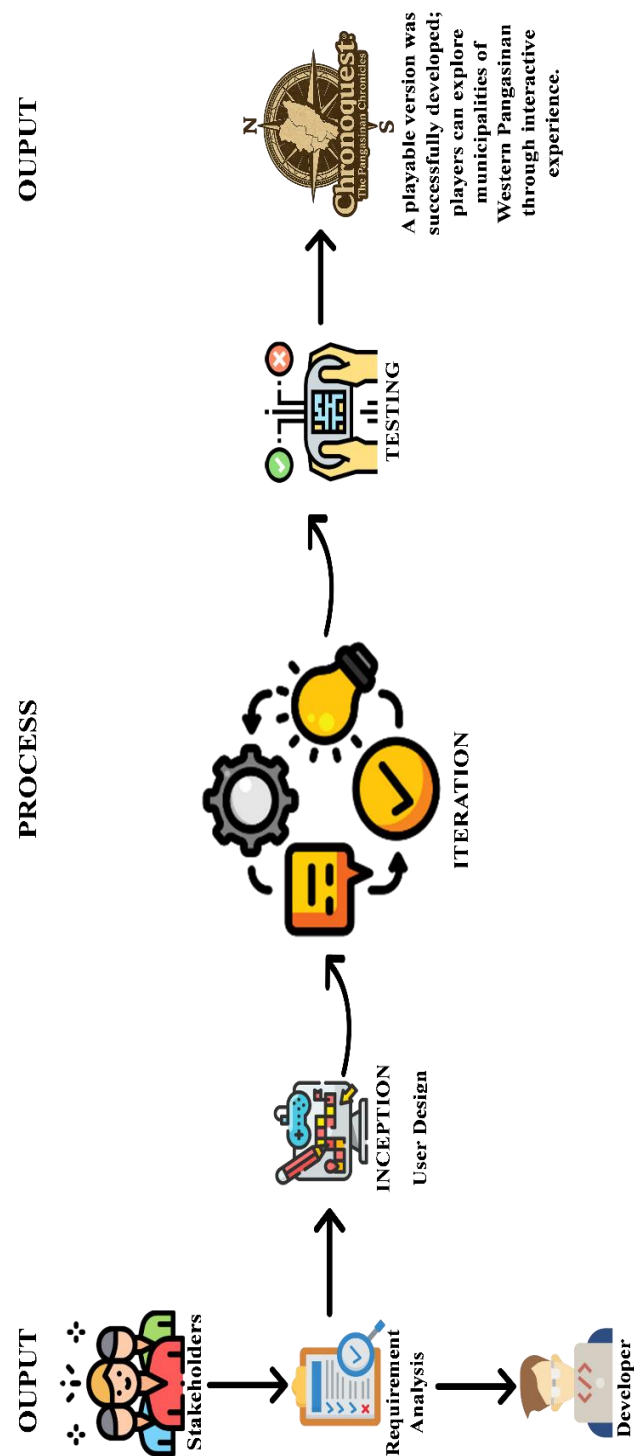


Figure 1. Input Process Output Process



Statement of Objectives The main objective of this study is to develop a historical role-playing game entitled *Chronoquest: The Pangasinan Chronicles*.

Specifically, the study seeks to accomplish the following objectives:

1. To identify the existing process in *Chronoquest: The Pangasinan Chronicles*;
2. To describe the features of the proposed *Chronoquest: The Pangasinan Chronicles*; and
3. To test the usability of the *Chronoquest: The Pangasinan Chronicles*.

Significance of the Study

The developers aimed to develop the *Chronoquest: The Pangasinan Chronicles* to provide enjoyable application software that would help the player to remember the most significant history in the Western Pangasinan.

The findings of this study are expected to contribute to the following:

Teachers. By utilizing *Chronoquest: The Pangasinan Chronicles* in their teaching methods, teachers can create dynamic learning environments that capture interest and improve retention of historical content, it will allow students to actively participate in their history subject discussion specifically Bolinao and Bani in Western Pangasinan. While history books remain the foundation of learning, this game offers an engaging visual and interactive tool that helps students better understand and remember local historical events.



Students. The game provides an enjoyable and immersive way to learn about the rich history and culture of Western Pangasinan. Aims to help students remember what they have learnt, turning history rote memorization to fun and engaging familiarization. Learners can experience the past through quest, characters, and in game challenges/missions that promote critical thinking and comprehension. Through gameplay, students can develop curiosity, analytical thinking, and a deeper connection to their cultural roots while having fun learning.

Tourism. The game is expected to showcase the rich cultural heritage and history of Western Pangasinan which can capture the interest of other people especially those curious **about the historical events that happened in Western Pangasinan.**

Future Developers and IT students. By means of this study, it aims to inspire future IT developers to create history-based game to help preserve history since most of our cultural heritage and community history are slowly forgotten by younger generation as they engage more in online games. The study could be a source of information and would be the foundation of a new and improved platform.

Scope and Delimitation of the Study

This study focuses on the development and design of *Chronoquest: The Pangasinan Chronicles*, a history-based educational game intended to provide an interactive learning experience while preserving local history.



The game allows players to explore historical narratives through quests, character interactions, quizzes, and cutscenes. It is designed primarily for educational and engagement purposes, particularly for students and young learners who wish to study history in a more immersive way.

The scope of the game is limited to two municipalities in Western Pangasinan: Bolinao and Bani. Players may choose only one municipality to explore at a time, and there is no chapter selection feature. Each municipality follows a linear sequence of quests that must be completed in order. In Bolinao, gameplay consists of three main quests: (1) finding the Bolinao Skull with gold teeth by exploring the environment and interacting with non-player characters (NPCs); (2) learning about the First Mass in Bolinao by entering the church and engaging in dialogues with characters to complete missions; and (3) answering a quiz based on the previous quests, followed by a cutscene highlighting Bolinao's history.

In Bani, the game begins with a short introduction to the municipality. The first quest centers on the legend behind the origin of the name "Bani," requiring players to interact with NPCs to progress. The second quest involves assisting local farmers in harvesting pakwan and rice, reflecting the municipality's agricultural background. After completing the quests, players answer a quiz, followed by a cutscene summarizing Bani's historical background. The study is limited to the development and usability of these two municipalities only.

The game does not include additional towns, open-world exploration beyond the designated quest areas, multiplayer features, or advanced difficulty



levels. Historical content is based on selected narratives and is presented in simplified form to support learning and engagement rather than relying on text-heavy rote memorization.

The game is designed as a single-player experience and focuses solely on the historical narratives of Bolinao and Bani. It is compatible with Android mobile devices and can be played offline, ensuring accessibility without requiring an internet connection.

This study defines clear boundaries for the project: *Chronoquest* is committed to its primary goal of delivering a history-rich, story-driven experience through municipality-based exploration. Its core objective is to make local history both accessible and engaging through interactive storytelling.

Definition of Terms

3D. Stands for three-dimensional. Everything we see in the world around us is 3D or exists in three dimensions. These dimensions are: 1) height, 2) width and 3) depth (length)

Artifacts. Objects made or used by humans that have historical or cultural significance. Artifacts appear as in-game items that players discover.

Chronicles. A record or narrative description of past events.

Chronological Order. Arranging events or items in the sequence they happened or occurred, starting with the earliest and ending with the most recent.



Chrono Quests. Are timed missions given at the start of a game, and once complete they reward an item that allows you to craft special equipment, generally the best in the game.

Game-Based Learning. An approach that uses games to achieve educational objectives. Chronoquest uses quests, storytelling, and quizzes to teach history in an engaging and interactive way.

History-Based Game. Those games that in some way represent the past, relate to discussions about it, or stimulate practices related to history.

Interactive. Means involving two-way communication or action between a user and a system, device, or another person, it allows both sides to respond to each other in real time.

NPC (non-Player Character). A character in a game controlled by the system, not by the player. NPCs guide players, provide quests, share historical information, and represent community members and local heroes.

Quest. To spend a long time searching for something that is difficult to find, or to spend a long time trying to achieve something difficult.

Single Player. A game mode where only one player plays independently. Chronoquest is a single-player game, allowing users to explore history at their own pace without competition.

Software. A set of programs and data that tell a computer or device how to perform tasks. Chronoquest is a software application that delivers historical content through interactive gameplay.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related literature and studies that served as the foundation for this research. The study aims to provide an overview of existing knowledge on the topic *Chronoquest: The Western Pangasinan Chronicles*, identify areas requiring further investigation, and establish a theoretical framework for the research.

The Existing Processes in History-Based Games

Abante (2020) highlighted that traditional games in barangays of Goa, Camarines Sur, adjacent to Mt. Isarog, reflect deep cultural and historical values that serve as tools for preserving local heritage. The study emphasized that games can transmit culture and history, particularly to younger generations who are losing touch with their roots.

Similarly, Abidin (2023) revealed the impact of history games in the Philippines, noting that interactive platforms provide students with engaging and less stressful environments. This approach led to higher memory retention, especially when learners actively participated in discussions of historical events.

Docuyanan et al. (2024) developed *Historya*, an interactive 3D firstperson shooter game about the Bataan Death March. Their project demonstrated how digital games can effectively teach Philippine history, enhancing student participation, curiosity, and cultural appreciation.



Amando and Llaneta (2024) discussed challenges in Philippine history education in the age of disinformation, where students are exposed to unreliable online content. They suggested that innovative methods, such as digital and interactive media, can enhance historical awareness and critical thinking.

Banerjee (2024) argued that integrating digital tools in teaching history makes learning more interactive and effective compared to traditional methods. The study found that technology-based environments sustain students' attention and engagement.

Bjorner (2024) further supported this view, showing that game-based learning significantly improves students' interest and retention in history education. Digital games encouraged learners to explore historical events and understand cultural contexts more deeply.

Hellerstedt and Mozeliuss (2024) examined student perceptions of gamebased learning for history. Using surveys and interviews, they found that students preferred interactive methods because they promoted enjoyment and deeper understanding of past events.

Fahrudin and Saefudin (2025) emphasized the use of primary sources in online history learning to enhance engagement and retention. Their study found that integrating authentic materials into digital games allows learners to connect more meaningfully with historical narratives.

Lee and Park (2021) noted that game-based learning not only improves student performance but also provides opportunities for assessment through challenges and missions integrated into gameplay.



Liu and Zhang (2024) discussed the application of digital game-based learning to promote cultural and historical heritage. Their findings revealed that history-themed games strengthen cultural identity and appreciation among players through interactive storytelling.

Existing studies consistently demonstrate the effectiveness of gamebased learning in history education. Researchers across different contexts have shown that integrating historical content into interactive games transforms traditional learning into a dynamic and immersive experience. These studies highlight that history-based games enhance cognitive learning, foster cultural appreciation, and create emotional connections to historical events.

However, most existing history-based games focus on national or global events rather than local heritage and culture. This gap underscores the need for educational tools that highlight regional histories.

Chronoquest: The Pangasinan Chronicles addresses this gap by developing an interactive game that presents the history of Western Pangasinan's municipalities through immersive gameplay. By incorporating quests, missions, and storytelling elements, the game aims to make history engaging and meaningful for learners. This study supports the idea that gamebased learning can be an effective way to teach and preserve local history, while also promoting cultural identity and appreciation among students.



Features of History-Based Games

Centeno et al. (2024) developed *KaSAYSAYan: The Integration of Philippine History as an Interactive and Informative Mobile Game Application* using the Unity Game Engine. The mobile game integrates Philippine history into an engaging experience through features such as interactive storytelling, quiz-based challenges, and artifact collection. These elements promote not only cognitive understanding of history but also emotional and cultural connections to Filipino heritage.

Liu and Zhang (2024) examined digital game-based learning for historical and cultural heritage in China. Their study emphasized immersive exploration of historical events, where players interact with historical characters and environments. Features such as story-driven gameplay, interactive timelines, and scenario-based problem solving were found to enhance engagement and improve retention of complex historical concepts.

Mendoza, Franco, and Gomez (2023) highlighted the role of technology in preserving cultural heritage through games. Features such as 3D modeling and interactive storytelling allow players to explore and manipulate historical sites and artifacts, creating rich educational experiences. This approach also improves accessibility to cultural heritage, particularly for students unable to visit historical sites physically.

Moscoso et al. (2020) demonstrated that immersive and gamified learning enables students to experience Philippine history firsthand, such as reenacting the 1896 revolution. The game incorporated role-playing mechanics,



decision-making scenarios, and historically accurate events, promoting critical thinking and empathy toward historical figures and contexts.

Pamintuan (2021) explored cultural motifs in games like *Anito: Paglalaro sa Lunan ng mga Arketipo at Laylayan*. Features included narrated-driven quests, symbolic representations of deities, and moral decision-making.

These elements strengthened cultural literacy while maintaining engagement through playful exploration.

Rivadulla et al. (2017) developed *PhilLoveHistory*, a mobile game designed to enhance knowledge of Philippine history. Key features included mini-games, historical fact challenges, and interactive storytelling. These features encouraged repeated engagement and self-paced learning while integrating local cultural contexts for relevance.

Sousa et al. (2024) emphasized adaptive learning in history-based games. Features such as personalized feedback, multiple narrative paths, and scenario branching allowed difficulty levels to adjust according to player progress. This adaptability enhanced learning outcomes and replayability, ensuring long-term engagement.

Triyoga (2024) highlighted the effectiveness of educational games like QuizWhizzer-based history applications. Features such as interactive quizzes, leaderboards, and immediate feedback provided motivation and measurable learning progress, reinforcing historical knowledge retention through gamification.



William et al. (2024) created *History Leap*, a game-based approach blending narrative-driven missions with historical exploration. Features included time travel mechanics, artifact collection, and scenario-based challenges, which allowed students to actively engage with historical events while fostering both cognitive understanding and emotional connection.

Zirawaga et al. (2017) observed that educational games support history teaching by combining problem-solving tasks, strategic decision-making, and interactive simulations. Features such as scenario branching, adaptive difficulty, and historical role-playing encouraged learners to construct knowledge actively rather than relying on passive memorization.

Across these studies, history-based games consistently demonstrate the value of interactive and immersive features in enhancing learning and cultural preservation. Common features include interactive storytelling, 3D simulations, role-playing mechanics, narrative-driven quests, adaptive learning, and gamification elements such as quizzes and leaderboards. These features promote both cognitive and emotional engagement, transforming history education from passive memorization into active exploration.

Overall, history-based games effectively preserve culture, improve historical literacy, and foster deeper connections to heritage by making learning both meaningful and enjoyable.



Usability of History-Based Games

Rodrigo et al. (2016) tested the mobile augmented reality game *Igpaw: Intramuros* with high school and college students (ages 16–22) at Ateneo de Manila University. Users found the game informative and engaging, with positive feedback on immersive storytelling. The study reported improvements in historical knowledge and cultural awareness.

Bjørner (2021) conducted a systematic review of game-based learning (GBL) in teaching history across K–12 education. Most studies reported positive effects on student motivation and engagement, though measurable learning outcomes were less consistent. Usability was generally good but highly context-dependent, varying with the type of game and classroom setting.

Watson et al. (2020) examined the video game *Making History* in a sophomore high school class in the USA. Students experienced enhanced engagement and improved critical thinking skills. Usability feedback indicated that the game was accessible, motivating, and contributed to better understanding of historical processes and decision-making.

Ortega et al. (2024) conducted a usability study on *Yamashita's Treasure*, a game-based instructional material for teaching Philippine history to secondary students. The game obtained high System Usability Scale (SUS) scores, with users finding it easy and enjoyable to use. Results showed increased student interest and improved knowledge retention.

Smith and Johnson (2018) evaluated *TimeQuest*, a PC game designed to teach medieval European history to UK high school students. The study found



the game user-friendly, with intuitive controls and engaging content. Students appreciated the historical accuracy and interactive storytelling, though some noted technical glitches on older hardware.

Centeno et al. (2023) conducted usability testing on *KaSAYSAYan: The Integration of Philippine History as an Interactive and Informative Mobile Game Application*. Results revealed high usability among Grade 8 students, who found the interface easy to navigate, instructions clear, and gameplay engaging. Trivia-based challenges helped learners better understand historical topics, making the experience more enjoyable than traditional methods.

Martínez et al. (2022) investigated the usability of *IncaQuest*, a mobile game teaching Incan history to Peruvian children aged 10–14. The study reported high engagement and ease of navigation, with children particularly enjoying mini-games. Some usability feedback suggested interface improvements for younger users.

Chen and Liu (2020) explored the usability of *Dynasty Chronicles*, a strategy game teaching Chinese history among university students in Taiwan. The interface was praised for clarity and responsiveness, and users appreciated contextual historical scenarios. However, some noted a steep learning curve for beginners.

Pérez et al. (2017) studied *Colonial Encounters*, a web-based game teaching Latin American colonial history to secondary students in Chile. Students found the game easy to use and enjoyed the narrative approach, though some reported difficulties with navigation and unclear instructions in certain modules.



Ahmed and Saleh (2023) assessed *Pharaoh's Legacy*, a historical puzzle game about ancient Egypt, with middle school students. The game scored well on usability tests for its simple interface and clear instructions, boosting curiosity and improving historical knowledge retention.

Synthesis and Research Gap

The review of related literature and studies reveals that game-based learning is a powerful tool for enhancing student motivation, engagement, retention, and critical thinking in history education. Foreign studies emphasize the role of interactive, digital, and role-playing history games in increasing student comprehension and motivation, while local studies highlight the impact of interactive storytelling and gamified historical experiences on emotional connection and cultural preservation.

Both foreign and local research consistently support game-based learning as an effective means of making history education more engaging and immersive. However, most studies have focused on national history or broad global themes, with limited attention to municipality-based historical narratives such as those found in Western Pangasinan. Local studies emphasize gamification's role in Philippine schools, while foreign studies highlight the effectiveness of interactive digital platforms in historical learning. Despite these contributions, there remains limited application of game-based learning in teaching regional and local history in the Philippines.

CHAPTER 3

DESIGN AND METHODOLOGY

This chapter presents the research methods used in the study. It explains the research design selected, the rationale behind its use, the tools employed for data collection, the process followed during the study, and the approach for analyzing and interpreting the data.

Research Design

According to Barroga et al. (2019), quantitative research involves the systematic collection and analysis of numerical data to explain, predict, or control phenomena through objective measurement and statistical analysis.

In this study, the developers adopted a quantitative research design. This approach was chosen because it focuses on collecting and analyzing measurable data, particularly through survey results. Quantitative methods allowed the researchers to obtain numerical information that could be analyzed using percentages, thereby identifying patterns and trends related to students' learning experiences and engagement with Western Pangasinan history.

By employing this design, the study was able to generate objective insights into how learners respond to history-based games. The use of numerical data ensured reliability and provided a strong basis for evaluating the effectiveness of *Chronoquest: The Pangasinan Chronicles* as an educational tool.



For data collection, a survey questionnaire was used to gather information from respondents. The questionnaire was carefully designed to encourage honest and accurate responses. It included items that measured usability, engagement, and perceived learning outcomes.

This method helped the researchers capture participants' opinions, experiences, and insights more clearly. The data collected provided reliable evidence to support the findings of the study and served as a foundation for refining the design and usability of *Chronoquest: The Pangasinan Chronicles*.

The study employed a quantitative research design supported by survey questionnaires to collect measurable data. This approach was appropriate because it allowed the researchers to analyze numerical results, identify trends, and evaluate the usability and educational impact of the game. Through this methodology, the study was able to assess how interactive, municipality-based history games can enhance student engagement and appreciation of Western Pangasinan's heritage.

Software Development Model

The Agile Development methodology was used as the guiding framework in this study. Agile is considered one of the best approaches for game development because of its iterative and incremental nature, which allows teams to deliver improved versions of the game frequently. This enables continuous refinement, testing, and user collaboration (Madampe & Hoda, 2020). The methodology consists of six phases: concept, inception, iteration, testing, release, maintenance, and review.



The developers adopted the Agile Model as the software development life cycle (SDLC) framework in creating *Chronoquest: The Pangasinan Chronicles*. Agile was chosen due to its focus on iterative development, prototyping, and user feedback. Since the game features municipalities in Western Pangasinan, Agile's sprint-based process was particularly well-suited, allowing each area to be developed, tested, and refined separately.

This approach promoted collaboration and continuous improvement, ensuring that the game evolved step by step throughout the development process. It also guaranteed that the final product was not only historically engaging but also aligned with the needs and preferences of its target users. By applying Agile principles, the developers were able to integrate user-centered design, iterative prototyping, and responsive adjustments, resulting in a more effective and meaningful educational game.

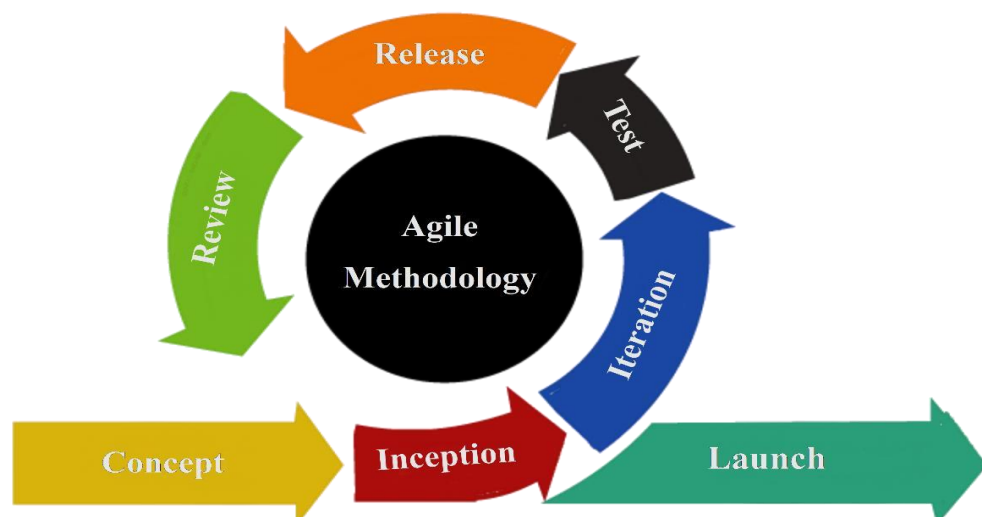


Figure 3. SDLC Model Diagram- The Agile Model



Phases of the Agile Life Cycle

Concept. This phase marks the beginning of development for *Chronoquest: The Pangasinan Chronicles*. The developers conducted surveys with target users to identify design preferences and gameplay features that would make the experience engaging and informative. From these findings, the team outlined initial requirements such as storytelling structure, art style, and gameplay mechanics. At this stage, requirements were kept minimal and flexible to allow adjustments as new ideas and user needs emerged. This ensured that the foundation of *Chronoquest* was realistic, achievable, and aligned with user expectations.

Inception. Once the concept was defined, the developers collaborated to establish the vision of presenting Western Pangasinan's history through interactive gameplay. During this phase, the team created early prototypes of the game's interface, navigation, and visual style. The game's architecture was established, including the narrative structure for municipalities such as Bolinao and Bani, ensuring that historical references and resources were integrated into the design.

Iteration. The iteration phase, also known as the construction phase, was the longest and most critical stage. Developers worked collaboratively to integrate historical research, cultural elements, and gameplay mechanics into the project, turning design concepts into functional code. Each sprint focused on specific features, such as developing one municipality's storyline, designing



hero characters, or creating interactive quests that showcased Western Pangasinan's heritage. The first playable version featured Bolinao, demonstrating the game's core functionality. Subsequent sprints added additional municipalities, with testing conducted after each iteration to ensure proper functionality and error-free performance. This process continued until all featured municipalities were tested, reviewed, and ready for release.

Release. At this stage, the game was nearly ready for deployment. Playtesting was conducted to ensure stability, eliminate bugs, and confirm that historical narratives functioned as intended. Each municipality's storyline, character interactions, and gameplay mechanics were carefully tested to provide a smooth and immersive experience. Game manuals were prepared to guide users before playing. After several rounds of testing and refinement, the final polished version of *Chronoquest: The Pangasinan Chronicles* was released to players.

Testing. Once the game was deployed and available to players, it entered the testing phase in real-world conditions. Developers closely observed gameplay, monitored performance, and collected feedback from users. This phase focused on identifying bugs, crashes, or confusing elements encountered by players, ensuring that the game remained functional and user-friendly.

Review. The review phase involved evaluating the game from a broader perspective. Developers gathered long-term feedback, analyzed gameplay trends, and assessed whether the game continued to meet modern device standards and player expectations. This phase also allowed the team to reflect



on successes and areas for improvement, providing insights for potential future versions or updates of the game.

Sources of Data

In this study, the developers utilized two primary sources of data. The first source was gathered through a survey consisting of ten questions, distributed to 100 respondents, most of whom were students. The survey aimed to determine their background knowledge, gameplay preferences, and expectations in playing an engaging and educational history game. Insights gained from this primary data helped shape the game's design, features, and narrative flow.

The second source of data came from online research. To ensure accuracy and credibility, the developers carefully examined the reliability and legitimacy of the digital resources consulted. Priority was given to updated and relevant materials related to history-based learning, game design, and the historical background of Western Pangasinan. This secondary data provided the foundation for integrating authentic historical and cultural elements into the game.

By combining survey results with reliable online research, the developers gathered comprehensive information that strengthened the foundation of *Chronoquest: The Pangasinan Chronicles*. This multi-source approach provided depth, reliability, and validity, guiding the development



process and ensuring that the game aligns with both educational and entertainment objectives.

Instrumentation and Data Collection

This section explains how the developers gathered data to ensure accuracy and authenticity in the study.

Survey Questionnaires. A survey questionnaire is a research instrument consisting of structured questions designed to collect data, opinions, or feedback from respondents, either through printed or online forms (Taherdoost, 2022). In this study, the developers conducted a survey to gather feedback from students about their interest and background experiences in learning history. The survey aimed to identify reasons behind their lack of engagement in the subject.

By interpreting the survey results, the developers determined common challenges faced by students, such as boredom, difficulty in retention, limited exposure, and lack of interactive learning materials. The information gathered served as a guide in designing *Chronoquest: The Pangasinan Chronicles*, ensuring that the game addresses these issues and provides an alternative tool for exploring Western Pangasinan history. This allowed users to learn while enjoying an interactive experience.

Online Research. Online research refers to the process of gathering information and data through internet-based tools and resources, allowing researchers to collect related materials quickly and efficiently for academic and



professional purposes (Synder, 2019). In this study, the developers explored multiple credible websites, articles, and related studies to gather accurate information.

Online research served as the foundation for creating the game by examining existing educational and historical games. This process allowed the developers to identify effective features, gameplay mechanics, and strategies that enhance learning and engagement. It also ensured that the historical content integrated into *Chronoquest* was authentic, culturally relevant, and educationally valuable.

Tools for Data Analysis

The developers used specific tools to collect and analyze data, serving as guidance for the creation of the proposed system. These tools helped in understanding related information and in shaping the development of *Chronoquest: The Pangasinan Chronicles*.

Flowchart. A flowchart is a diagram that shows the sequential steps of a process involved in the creation of a product or service. In this study, the developers used flowcharts to visualize and understand the flow of the game. This tool clarified the logical sequence of actions, decision points, and interactions, ensuring that the gameplay design was coherent and aligned with the intended learning objectives.

Gantt Chart. A Gantt chart is a project management tool that assists in planning and scheduling projects of all sizes. It is particularly useful for

visualizing timelines and task dependencies. In this study, the developers used Gantt charts to plan and manage the order of tasks, determine required resources, and monitor project progress. By converting tasks into horizontal bars, the chart provided a clear view of deadlines and responsibilities, ensuring that the development process was organized and completed within the expected timeframe.

Requirements Analysis

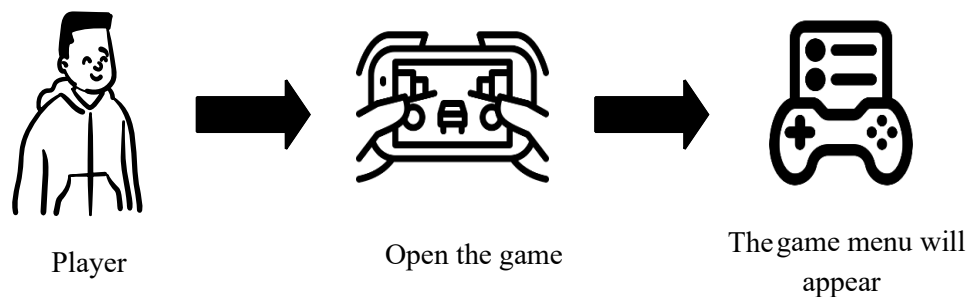


Figure 3.2 Opening the Android-based game

When the player opens the game, the main menu of the game appears.

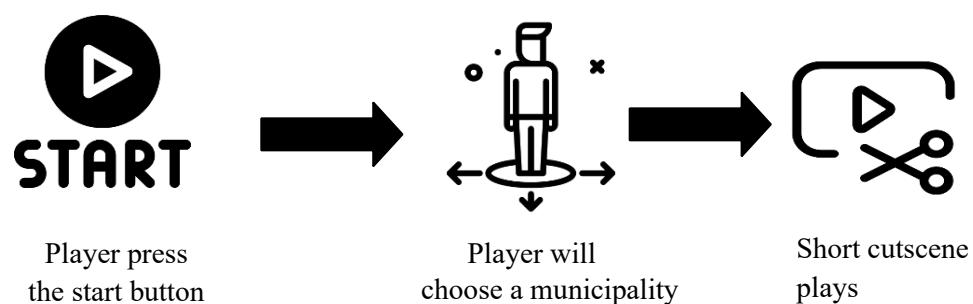


Figure 3.3 Selection of Municipality

After pressing the start button, the player sees a location selection screen where they choose a municipality to explore. Once a location is selected, a short cutscene plays.

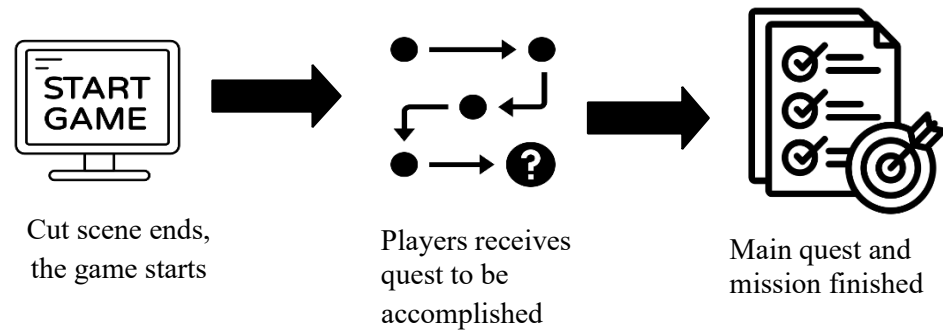
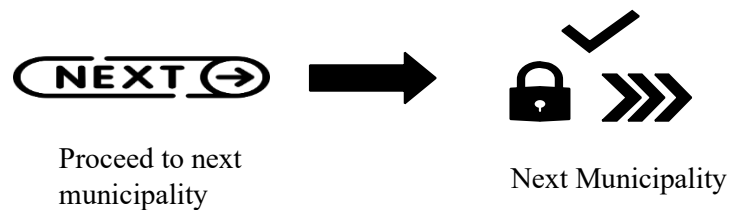


Figure 3.4 Game and Quest Starts

After selecting municipality, a short cutscene plays for historical context. Once the cutscene ends, the game begins, and the player receives a quest with missions to complete.



***Figure 3.5 Finishing Quests, Missions,
and Moving to Next Municipality***

After finishing all the main quests, it will proceed to the next municipality.

CHAPTER 4

DISCUSSION OF FINDINGS

The Existing Process

This chapter presents the findings of the study on the development of Chronoquest: The Pangasinan Chronicles. It includes the game flowchart, survey results from target users, and the problem encountered that motivated the creation of the game. Screenshots of the developed system are also provided, along with the results of usability testing to evaluate its functionality, efficiency, and user experience.

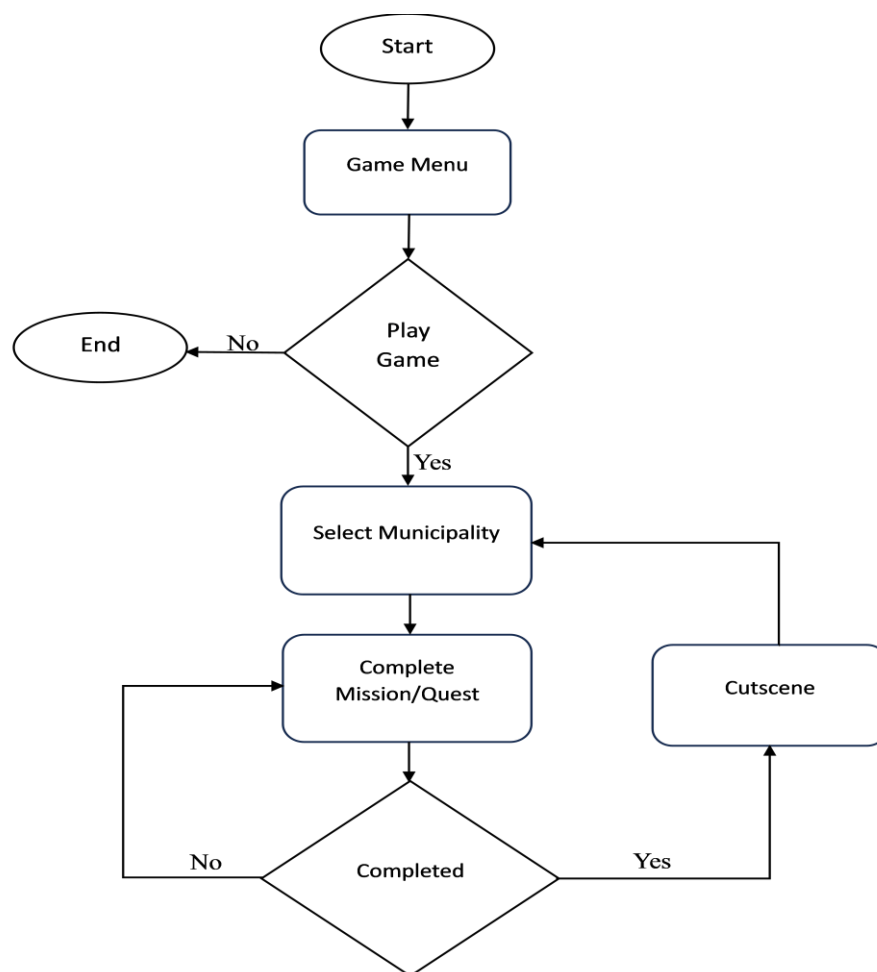


Figure 4. Graphical representation of the player's progress in the game



This chart illustrates the overall process of *Chronoquest: The Pangasinan Chronicles*, showing how the player progresses in every stage. When the player starts the game and already selected the first municipality in selection. The municipalities featured in the game is Bolinao and Bani located in Western Pangasinan. The flow continues until the game is completed.

The Developed System

System Design. The proposed system is designed to address the challenges faced by many students that are not aware or lack of knowledge about the history of Pangasinan. The developers will offer different experience of learning, allowing students to explore the history of 2 municipalities in Pangasinan in an interactive way, enhancing engagement, improving knowledge retention and developing critical thinking skills.

Survey Result

The following charts present the results of the data gathered from the respondents.

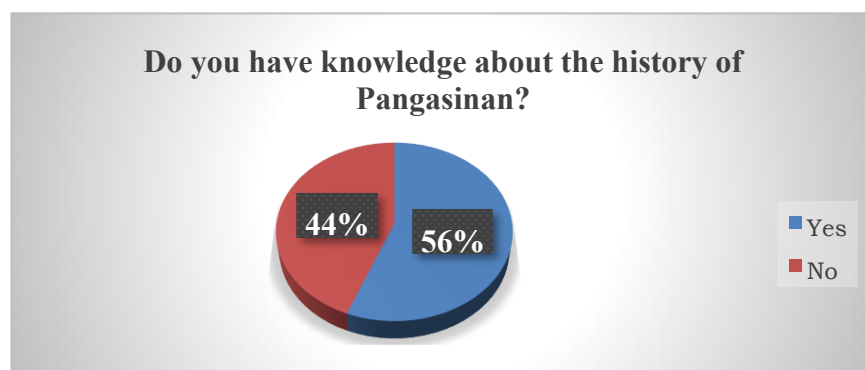


Figure 4.1 Survey Questionnaire #1



Out of the respondents, 56% stated that they have knowledge about the history of Pangasinan, while 44% admitted they do not. Interestingly, none of the participants chose "maybe," showing a clear divide in historical awareness.

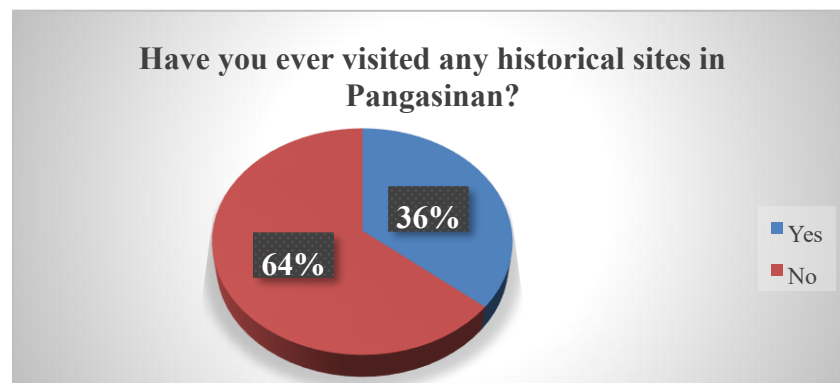


Figure 4.2 Survey Questionnaire #2

When asked if they have ever visited historical sites in Pangasinan, 36% answered "yes," while a larger portion, 64%, said "no," suggesting that many individuals may not have had the opportunity or interest to explore these locations.

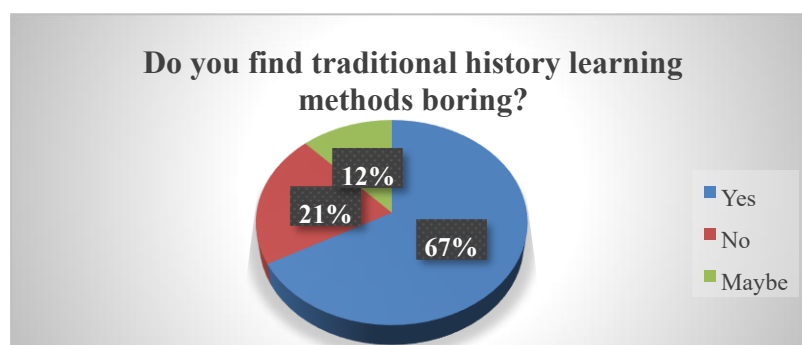


Figure 4.3 Survey Questionnaire #3



Regarding traditional history learning methods, 67% of respondents find them boring, 21% do not share this sentiment, and 12% are undecided. This indicates that a majority of learners are disengaged with conventional ways of teaching history.

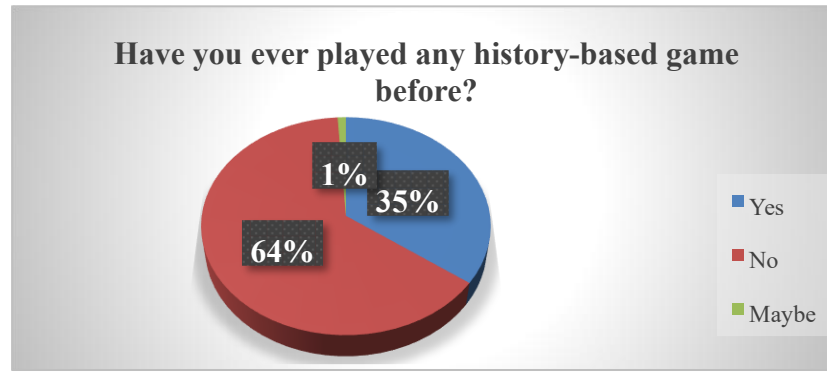


Figure 4.4 Survey Questionnaire #4

In terms of history-based games, 35% of the respondents have played one before, 64% have not, and 1% are unsure. This shows that while some individuals are familiar with educational gaming, most have yet to experience it.



Figure 4.5 Survey Questionnaire #5

For gaming devices, 96% of the respondents primarily use mobile phones, while only 4% play on PCs or laptops. None of them use gaming consoles, which highlights the dominance of mobile gaming among participants.

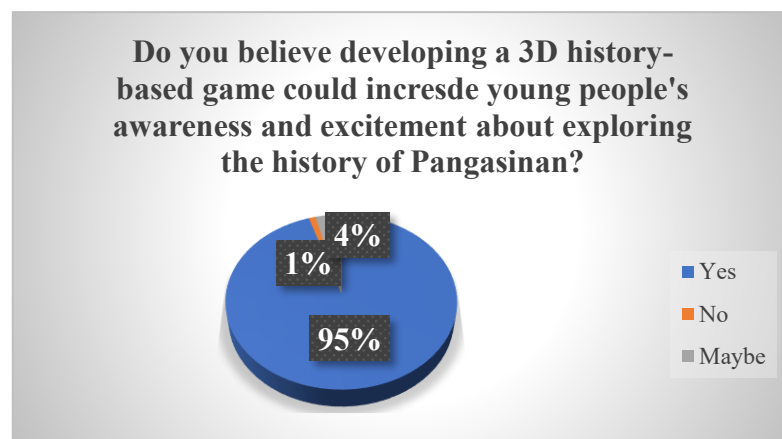


Figure 4.6 Survey Questionnaire #6

Out of 100 respondents, 95% of the respondents believe that developing a 3D history-based game could increase young people's awareness and excitement about the history of Pangasinan. Only 1% disagreed, while 4% remained uncertain.

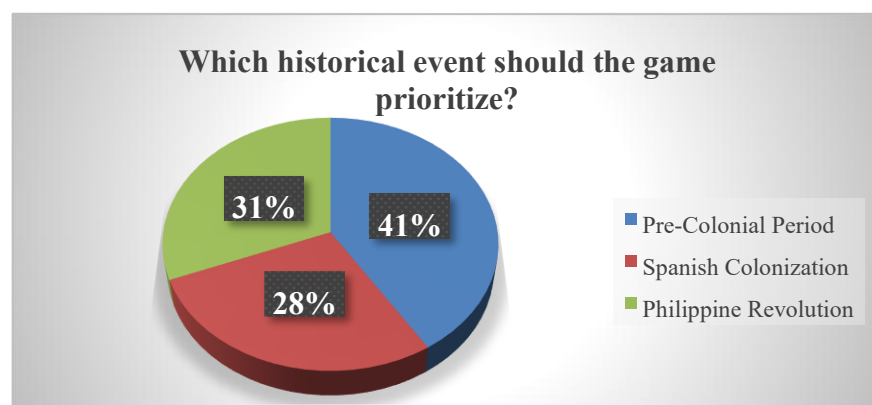


Figure 4.7 Survey Questionnaire #7

When asked which historical event the game should prioritize, 41% voted for the Pre-Colonial Period, 31% for Philippine Revolution period, and 28% for the Spanish Colonization. This indicates that most participants are more interested in the Pre-Colonial period.

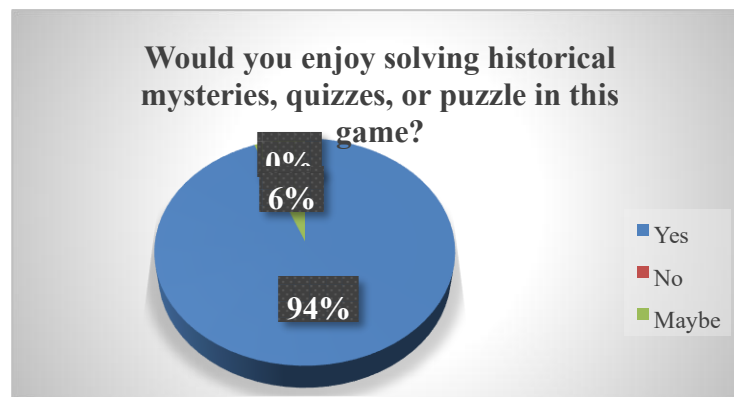


Figure 4.8 Survey Questionnaire #8

About 94% of respondents expressed interest in solving historical mysteries, quizzes, or puzzles within the game. None of them rejected the idea, and 6% were unsure.

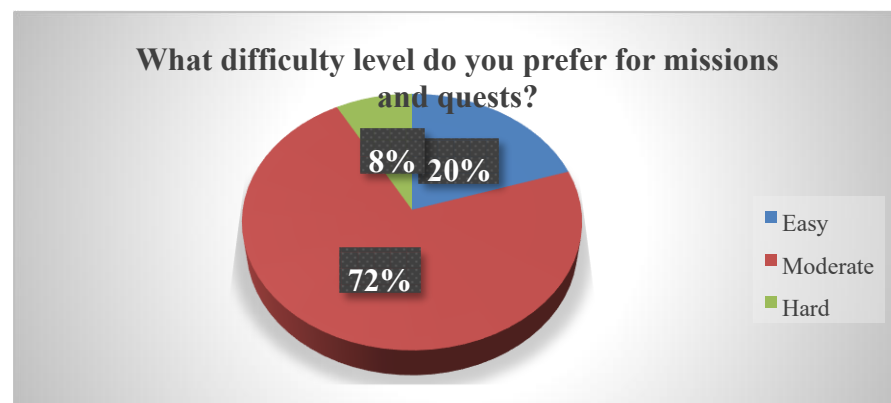


Figure 4.9 Survey Questionnaire #9

In terms of difficulty levels for missions and quests, 72% preferred a moderate level, 20% opted for an easy mode, and only 8% wanted a hard level. This reflects a desire for a balanced gaming experience that is neither too simple nor overly challenging.

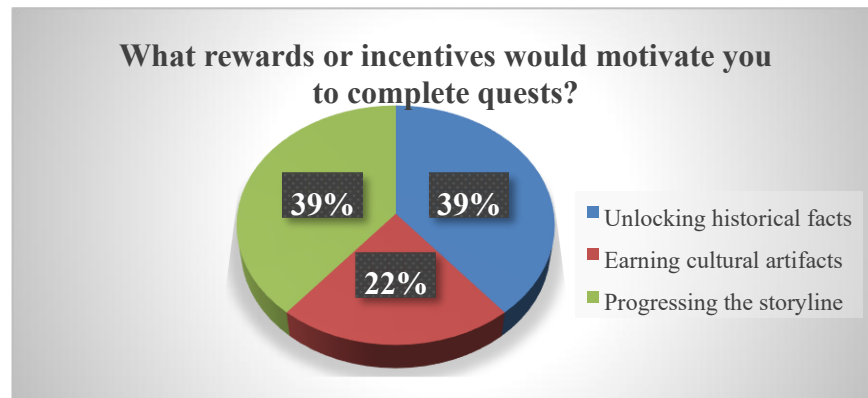


Figure 4.10 Survey Questionnaire #10

When it comes to rewards, 39% of respondents prefer unlocking historical facts, another 39% are motivated by progressing the storyline, and 22% like earning cultural artifacts. This shows that a mix of educational and narrative driven incentives would best engage players.

System Overview

This section presents the overview of the system.



Figure 4.11 Game Menu Display



The game menu serves as the main navigation screen of *Chronoquest: The Pangasinan Chronicles*. It allows players to start the game through the **Play** button, adjust preferences in **Setting**, view progress in **Achievements**, learn about the developers in **About Us**, and exit the game using **Exit**. The menu is designed to be simple and easy to navigate for users.



Figure 4.12 Municipality selection

The municipality selection screen allows players to choose municipality to explore after selecting Play. This feature gives players the freedom to decide between available municipalities, guiding them to different stories, quests, and historical content specific to each location.



Figure 4.13 Story Preview of Bolinao

The image shows the preview for the Bolinao municipality presents a short introduction to its first story, “Ang Nakatagong Gintong Bungo ng Bolinao.” This sets the historical atmosphere of the game by giving players a glimpse of the legend and events they will uncover, helping build interest and curiosity for players before starting the quest.

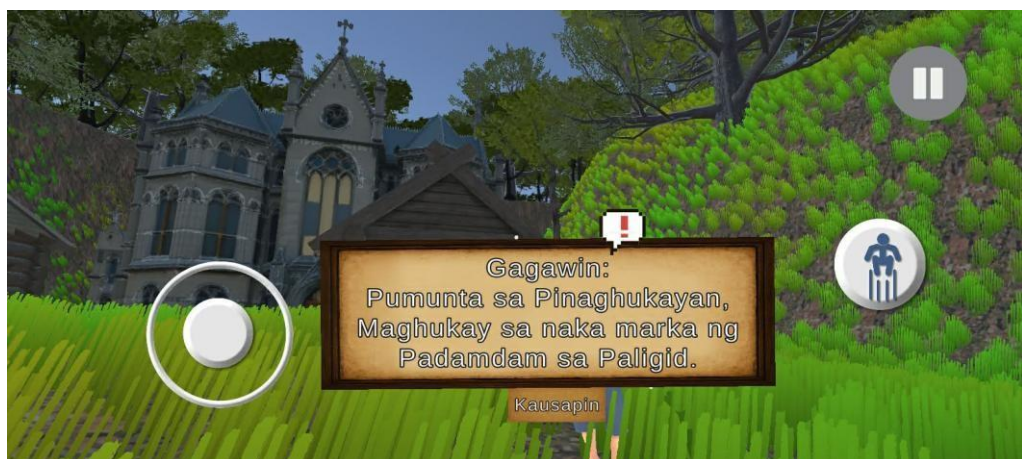


Figure 4.14 Bolinao Quest

The image presents the first quest in the Bolinao municipality, where players begin their adventure by exploring the environment using the control



buttons. Interaction with the community characters is required to unlock the quest, as the character provides instructions and historical context related to the task. This interaction guides the player on what actions to perform, emphasizing exploration and dialogue as key gameplay elements while introducing the historical narrative of Bolinao.



Figure 4.15 Church Environment in Bolinao

The image shows a church environment in Bolinao where players can freely explore the area. As the player approach the church, a text prompt appears displaying the location and name of the church. Although the structure is not an exact replica of the real Saint James the Great Parish in Bolinao, it is designed to give players a sense of place and encourage exploration while learning about the historical setting through the game environment.



Figure 4.16 Bolinao Quest

The image presents the second quest in the game, where the player is tasked with helping NPC characters accomplish a specific activity. Through interaction and cooperation with these characters, the player progresses in the quest while learning more about the historical context connected to the task, reinforcing engagement through active participation.



Figure 4.17 Story preview of Bani



The image shows the preview for the Bani municipality presents a short introduction to its first story, “Puno ng Himala”. It highlights the Bani tree and explains the origin and historical significance of the municipality’s name.

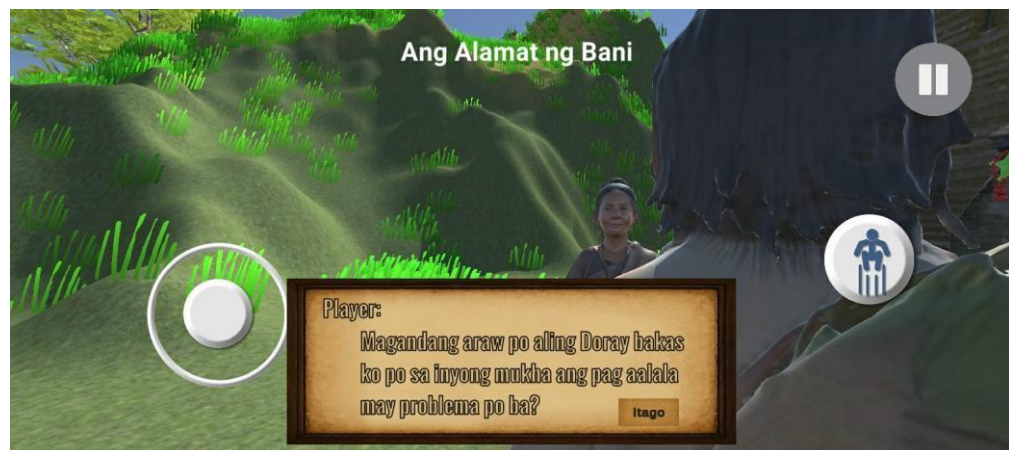


Figure 4.18 Interaction dialogues

The image shows an interaction dialogue between the player and community characters. These dialogues provide instructions, hints, and historical context, guiding the player through quests while making the learning experience more engaging and immersive.

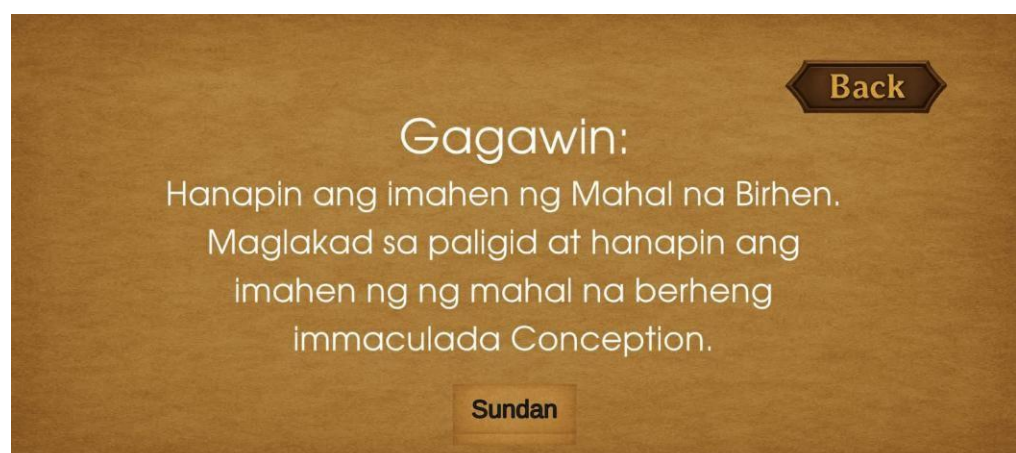




Figure 4.19 First Quest

The image shows a quest objectives in Bani where players receive missions from community characters. By completing these tasks, players actively engage with the game's objectives while learning about the culture and history of the municipality.



Figure 4.20 Second Quest

The image illustrates the second quest, where players will help community characters with tasks tied to the municipality, such as helping farmers, highlighting local livelihoods and encouraging players to connect with the community's and appreciate their roots.



Post-Test Results

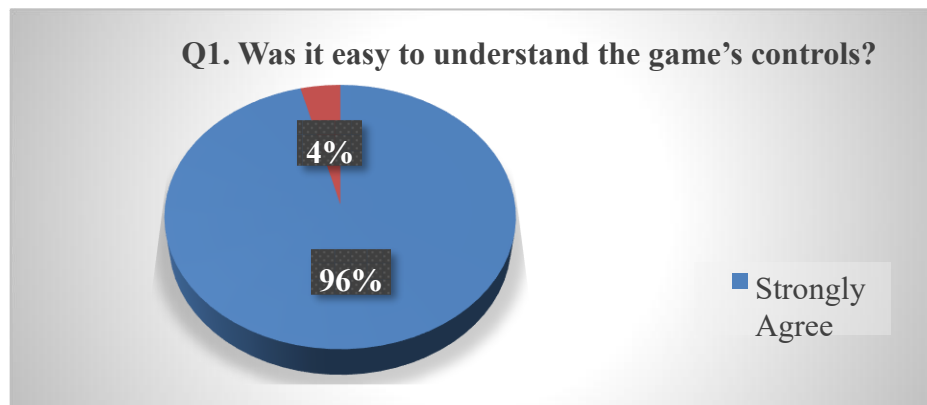


Figure 4.21 Question 1

The results show that players had no difficulty understanding the game controls, with 96% respondents strongly agreed and 4% agreed, indicating that the controls were simple and intuitive.

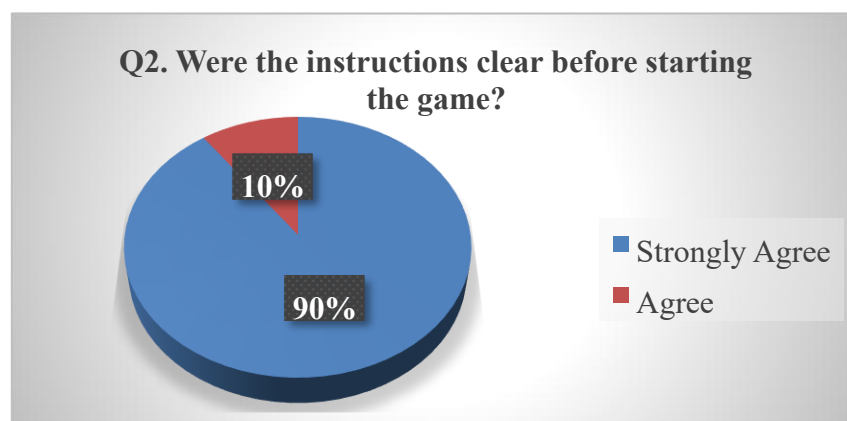


Figure 4.22 Question 2

All respondents found the instructions clear, as 90 strongly agreed and 10 agreed, suggesting that the game provided sufficient guidance before gameplay began.

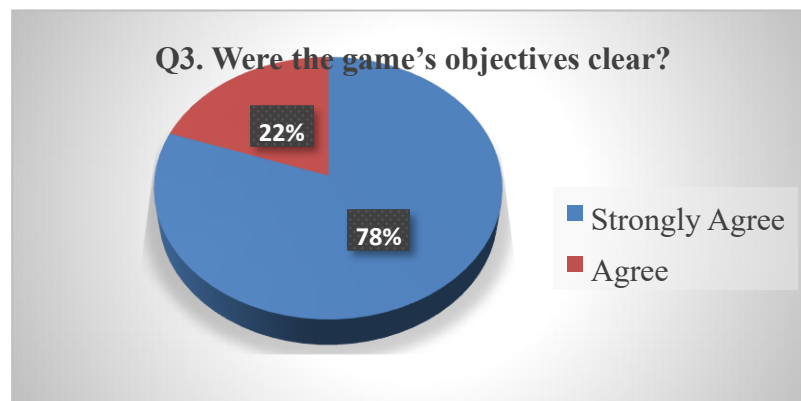


Figure 4.23 Question 3

Most players clearly understood the objectives of the game, with 78% strongly agreed and 19% agreed, showing that the goals were well explained and easy to follow.

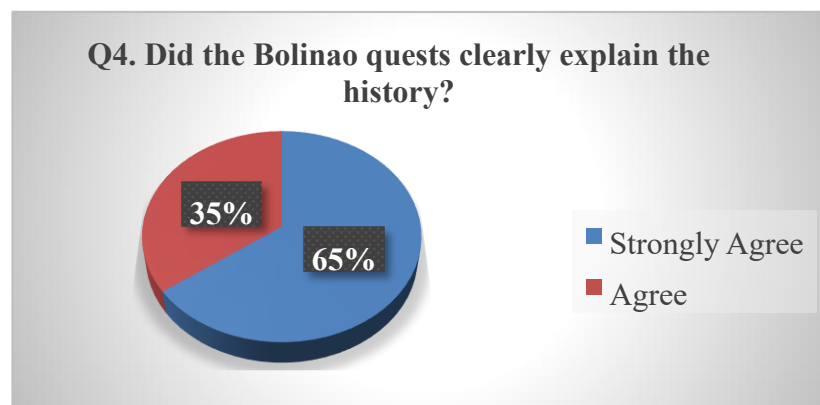


Figure 4.24 Question 4

The Bolinao quests were effective in explaining historical content, with 65% respondents strongly agreed and 35% agreed.

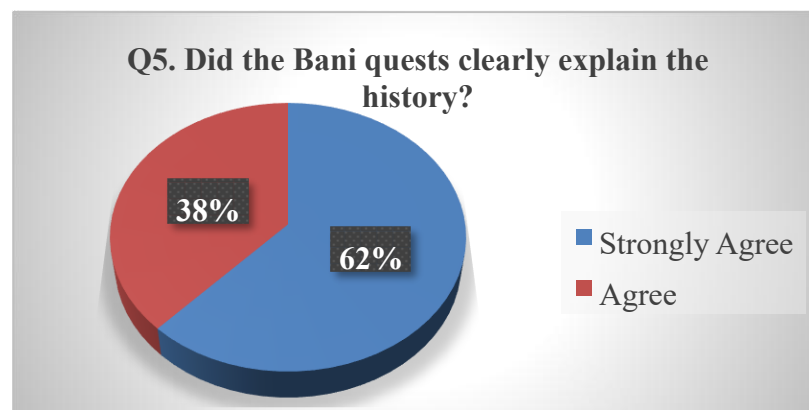


Figure 4.25 Question 5

The Bani quests successfully conveyed historical information, as 62% strongly agreed and 38% agreed.

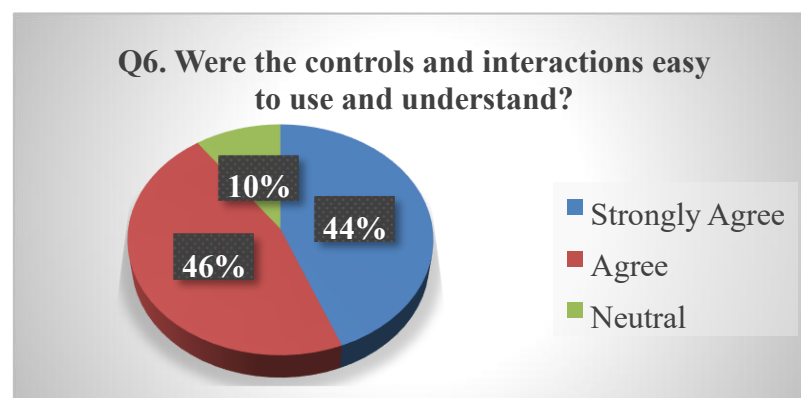


Figure 4.26 Question 6

Most respondents found the controls and interactions easy to use, with 44% strongly agreed and 46% agreed, while 10% remained neutral.

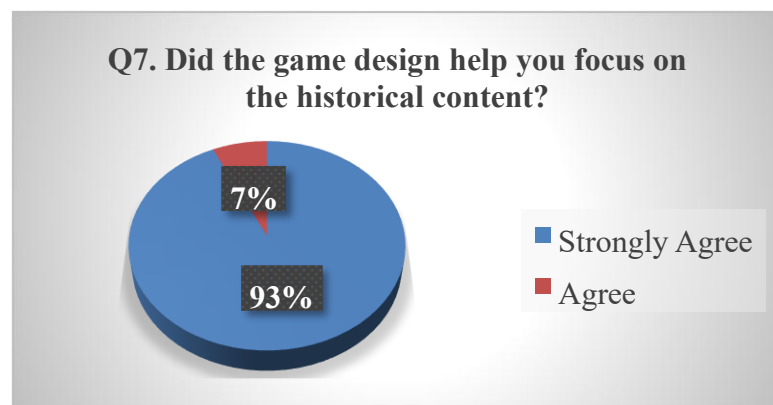


Figure 4.27 Question 7

The game design effectively guided players' attention toward historical learning, with 93% strongly agreed and 7% agreed.

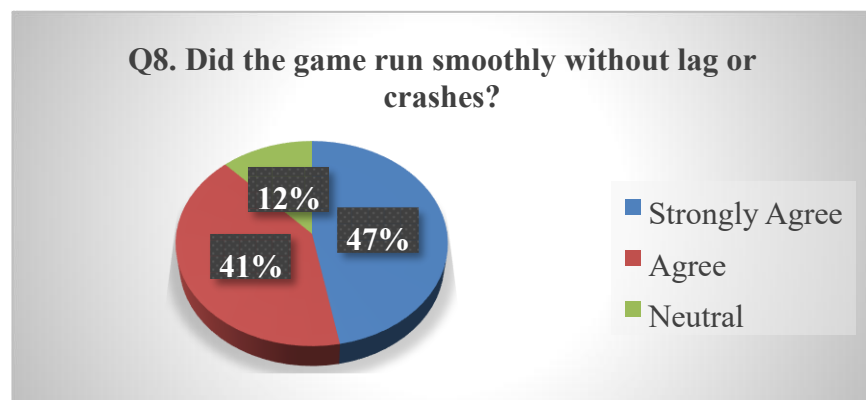


Figure 4.28 Question 8

The majority of players experienced smooth gameplay, with 47% strongly agreed and 41% agreed, though 12% respondents were neutral.

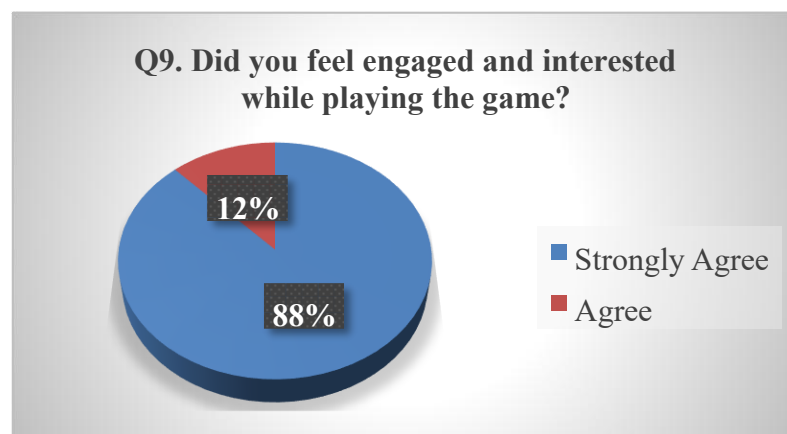




Figure 4.29 Question 9

A number of 88% respondents strongly agreed and 12% agreed, showing that the game was able to maintain interest throughout the experience.



Figure 4.30 Question 10

With 95% strongly agreed and 5% agreed, confirming that the game is both enjoyable and effective for learning history

CHAPTER 5

SUMMARY, CONCLUSION, RECOMMENDATION

Summary

This study focuses on the development of Chronoquest: The Pangasinan Chronicles, a history-based narrative game designed to address the challenges in learning and appreciating the history of two municipalities Western Pangasinan. The traditional approach to teaching history relies heavily on lectures and textbooks, which often result in low student engagement and poor retention of historical knowledge. Many young learners also lack exposure to interactive and innovative tools that could connect them to their cultural heritage in a more meaningful way.

The developers proposed an interactive digital game that integrates storytelling, puzzle-solving, and role-playing mechanics. The goal is to transform local history into an immersive experience that is both educational and entertaining. The development of the game is guided by the Agile Software Development Life Cycle (SDLC), which allows the system to be built through the phases of concept, inception, iteration, testing, release, and review. This methodology ensures flexibility, adaptability, and the opportunity to refine the game based on continuous feedback from playtests and surveys conducted among the target users. Key features of Chronoquest: The Pangasinan Chronicles include municipality-based that explore local historical narratives, interactive storytelling that brings historical events and folklore to life, and puzzles and quests that encourage active participation from players.



To attain the primary objective of this study, the developers specifically aimed to accomplish the following:

1. Conduct a survey to determine how users primarily students and local residents feel about learning history through interactive games.
2. Identify key features and design preferences based on the results of the conducted survey to guide the development of the game.
3. The developers will present a developed system of Chronoquest: The Pangasinan Chronicles to users in order to gather feedback on engagement, usability, and historical accuracy, as well as identify areas for improvement.

By incorporating these elements, the project aims to improve student engagement with history, enhance retention of historical knowledge, and promote awareness and appreciation of the cultural heritage of Western Pangasinan. Overall, this study emphasizes the development of a history-based educational game that addresses the lack of engaging learning tools for history education. By merging interactive gameplay with cultural and historical content, Chronoquest: The Pangasinan Chronicles seeks to contribute to both education and heritage preservation.

Conclusion

Based on the gathered findings and analysis of the developed game, the following conclusions were drawn:

1. Chronoquest: The Pangasinan Chronicles is a history-based game designed to be an engaging and interactive learning tool that fosters historical awareness, cultural appreciation, and curiosity among



students. It aims to deepen their understanding of Pangasinan's rich past through storytelling, exploration, and decision-making in a digital learning environment that complements traditional classroom instruction.

2. The challenges identified from the gathered data regarding Chronoquest: The Pangasinan Chronicles have been carefully considered and will guide further improvements to its gameplay, storyline, and educational features. These enhancements are intended to keep learners interested and motivated while strengthening their knowledge of local history and heritage.
3. The game's unique features have been recognized and will be optimized to help students grasp significant historical events, notable figures, and cultural landmarks of Pangasinan in a fun and meaningful way. By blending learning with entertainment, the game provides an enjoyable approach that helps maintain student engagement and understanding.

In conclusion, the development of Chronoquest: The Pangasinan Chronicles presents an innovative step toward integrating technology with historical education. It supports the advancement of digital learning, preserves local culture, and promotes critical thinking and historical literacy through immersive storytelling and interactive gameplay.



Recommendation

To further enhance gameplay quality, historical depth, and educational value, the following recommendations are proposed for the continued development and improvement of *Chronoquest: The Pangasinan Chronicles*:

Interactive Historical Quests. Add more missions or side quests inspired by real historical events, landmarks, and figures in Pangasinan to strengthen players' connection with the province's heritage.

Voice Narration and Local Dialect Support. Incorporate voiceovers and dialogues in both English and Pangasinan to preserve local language and make storytelling more authentic and engaging.

Character Customization. Allow players to personalize their avatars using traditional Pangasinense clothing and accessories to promote cultural identity and immersion.

Dynamic Map Exploration. Develop a more open or semi-open world that lets players freely explore Pangasinan's towns, heritage sites, and historical landmarks.

Achievement and Reward System. Include collectible items, unlockable achievements, or hidden artifacts that reveal additional historical facts when discovered.

Improved User Interface and Accessibility. Redesign menus and controls for better usability and add accessibility features such as subtitles, adjustable font sizes, and colorblind-friendly visuals.



Community Engagement Features. Introduce online leaderboards, discussion boards, or multiplayer cooperative missions where players can collaborate to complete historical challenges.

Cultural Mini-Games. Include mini-games based on traditional Pangasinense practices such as fishing, weaving, or local festivals to educate players about daily life and customs.

In-Game Encyclopedia. Introduce a “History Log” or “Chronopedia” where players can review information, artifacts, and biographies of key historical figures encountered during gameplay.

Periodic Content Updates. Schedule seasonal or historical event-based updates (like Hundred Islands Festival) to keep the game fresh and relevant.



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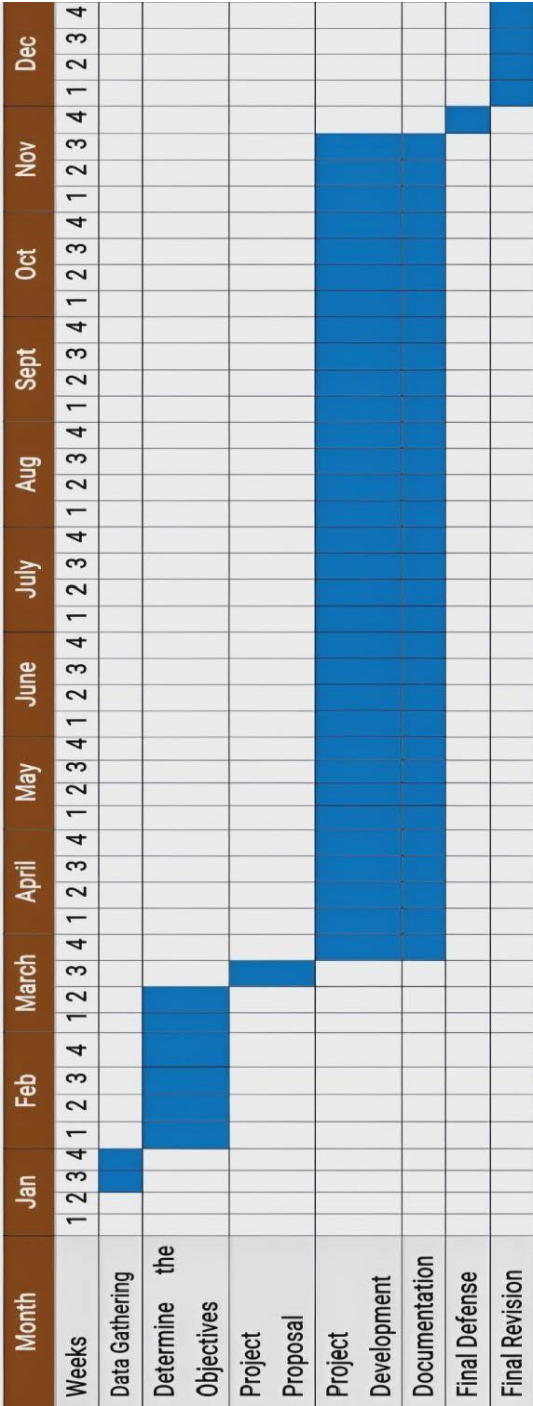
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APPENDIX A

GANNT CHART





APPENDIX B

PRE-TEST SURVEY QUESTIONNAIRES



Golden West Colleges
Chronoquest: The Pangasinan Chronicles



Name: _____ Date: _____

1. Do you have any knowledge about the history of Pangasinan?

☐ Yes ☐ No ☐ Maybe

2. Have you ever visited any historical sites in Pangasinan?

☐ Yes ☐ No

3. Do you find traditional history learning methods boring?

☐ Yes ☐ No ☐ Maybe

4. Have you ever played any history-based game before?

☐ Yes ☐ No ☐ Maybe

5. What technology devices you usually used in playing games?

☐ Mobile phones ☐ PC/laptops ☐ Console

Others: _____

6. Do you believe developing a 3D history-based game could increase young people's awareness and excitement about exploring the history of Pangasinan?

☐ Yes ☐ No ☐ Maybe

7. Which historical event should the game prioritize?

☐ Pre-Colonial period ☐ Spanish Colonization ☐ Pangasinan Revolution

Others: _____

8. Would you enjoy solving historical mysteries, quizzes or puzzles in this game?

☐ Yes ☐ No ☐ Maybe

9. What difficulty level do you prefer for missions and quest?

☐ Easy ☐ Moderate ☐ Hard

10. What rewards or incentives would motivate you to complete quests?

☐ Unlocking historical facts ☐ Earning cultural artifacts ☐ Progressing the storyline

Thank you for your participation

God Blessed



APPENDIX C

SAMPLE SURVEY


Golden West Colleges
Chronoquest: The Pangasinan Chronicles


Name: DE LEON, JASMIN ANN N. Date: 04-14-25

- Do you have any knowledge about the history of Pangasinan?
☒ Yes ☐ No ☐ Maybe
- Have you ever visited any historical sites in Pangasinan?
☒ Yes ☐ No
- Do you find traditional history learning methods boring?
☐ Yes ☒ No ☐ Maybe
- Have you ever played any history-based game before?
☐ Yes ☒ No ☐ Maybe
- What technology devices you usually used in playing games?
☒ Mobile phones ☐ PC/laptops ☐ Console
 Others: _____
- Do you believe developing a 3D history-based game could increase young people's awareness and excitement about exploring the history of Pangasinan?
☒ Yes ☐ No ☐ Maybe
- Which historical event should the game prioritize?
☐ Pre-Colonial period ☐ Spanish Colonization ☒ Pangasinan Revolution
 Others: _____
- Would you enjoy solving historical mysteries, quizzes or puzzles in this game?
☒ Yes ☐ No ☐ Maybe
- What difficulty level do you prefer for missions and quest?
☒ Easy ☐ Moderate ☐ Hard
- What rewards or incentives would motivate you to complete quests?
☐ Unlocking historical facts ☐ Earning cultural artifacts ☒ Progressing the storyline

Thank you for your participation
 God Blessed



Golden West Colleges
Chronoquest: The Pangasinan Chronicles



Name: CORPUZ RUBBIE G BSIT-SET-A Date: 04/14/25

1. Do you have any knowledge about the history of Pangasinan?
☒ Yes ☐ No ☐ Maybe
2. Have you ever visited any historical sites in Pangasinan?
☒ Yes ☐ No
3. Do you find traditional history learning methods boring?
☐ Yes ☐ No ☒ Maybe
4. Have you ever played any history-based game before?
☐ Yes ☒ No ☐ Maybe
5. What technology devices you usually used in playing games?
☒ Mobile phones ☐ PC/laptops ☐ Console
 Others: _____
6. Do you believe developing a 3D history-based game could increase young people's awareness and excitement about exploring the history of Pangasinan?
☒ Yes ☐ No ☐ Maybe
7. Which historical event should the game prioritize?
☒ Pre-Colonial period ☐ Spanish Colonization ☐ Pangasinan Revolution
 Others: _____
8. Would you enjoy solving historical mysteries, quizzes or puzzles in this game?
☒ Yes ☐ No ☐ Maybe
9. What difficulty level do you prefer for missions and quest?
☐ Easy ☒ Moderate ☐ Hard
10. What rewards or incentives would motivate you to complete quests?
☒ Unlocking historical facts ☐ Earning cultural artifacts ☐ Progressing the storyline

Thank you for your participation

God Blessed



APPENDIX D

POST-TEST SURVEY QUESTIONNAIRES

USABILITY	1	2	3	4	5
General Usability Questions:					
1. How easy was it to understand the game's controls?					
2. Were the instructions clear before starting the game?					
3. How clear were the game's objectives (e.g., interacting with characters, finding artifacts and complete missions)?					
Gameplay Experience Questions:					
4. Did the Bolinao quests (Bolinao Skull, First Mass, quiz, cutscene) clearly explain the history?					
5. Did the Bani quests (Origin of the name Bani, crops/farming, quiz, cutscene) clearly explain the history?					
6. Were the controls and interactions (movement, dialogue, quests) easy to use and understand?					
Interface and Visual and Design:					
7. Did the game design (graphics, environment, and layout) help you focus on the historical content?					
Performance and Technical Question:					
8. Did the game run smoothly without lag or crashes?					
Overall Usability and Satisfaction:					
9. Did you feel engaged and interested while playing the game?					
10. Overall, was your experience playing the game is it enjoyable and easy to use as a digital tool to learn history?					



APPENDIX E

SAMPLE SURVEY




Chronoquest: The Pangasinan Chronicles
4th YEAR BSIT

Overview: "*Chronoquest: The Pangasinan Chronicles*" is a history-based game designed to help players learn about the local history of Western Pangasinan through interactive storytelling, quest, and quizzes. The game features 2 municipalities Bolinao and Bani. Players must complete missions, interact with characters, and answer quizzes. The game aims to make learning history more engaging, interactive, and enjoyable for users.

Survey Evaluation Form for the Proposed System

5: Strongly Agree, 4: Agree, 3: Neutral, 2: Disagree, 1: Strongly Disagree

USABILITY	1	2	3	4	5
 General Usability Questions:					✓
1. How easy was it to understand the game's controls?					✓
2. Were the instructions clear before starting the game?					✓
3. How clear were the game's objectives (e.g., interacting with characters, finding artifacts and complete missions)?				✓	
 Gameplay Experience Questions:					
4. Did the Bolinao quests (Bolinao Skull, First Mass, quiz, cutscene) clearly explain the history?				✓	
5. Did the Bani quests (Origin of the name Bani, crops/farming, quiz, cutscene) clearly explain the history?				✓	
6. Were the controls and interactions (movement, dialogue, quests) easy to use and understand?					✓
 Interface and Visual and Design:					
7. Did the game design (graphics, environment, and layout) help you focus on the historical content?				✓	
 Performance and Technical Question:					
8. Did the game run smoothly without lag or crashes?			✓		
 Overall Usability and Satisfaction:					
9. Did you feel engaged and interested while playing the game?					✓
10. Overall, was your experience playing the game is it enjoyable and easy to use as a digital tool to learn history?					✓

Year level and set: Grade III - Zamora

Julian b. Nacipal

Signature over the Printed Name



Chronoquest: The Pangasinan Chronicles

4th YEAR BSIT

Overview: "Chronoquest: The Pangasinan Chronicles" is a history-based game designed to help players learn about the local history of Western Pangasinan through interactive storytelling, quest, and quizzes. The game features 2 municipalities Bolinao and Bani. Players must complete missions, interact with characters, and answer quizzes. The game aims to make learning history more engaging, interactive, and enjoyable for users.

Survey Evaluation Form for the Proposed System

5: Strongly Agree, 4: Agree, 3: Neutral, 2: Disagree, 1: Strongly Disagree

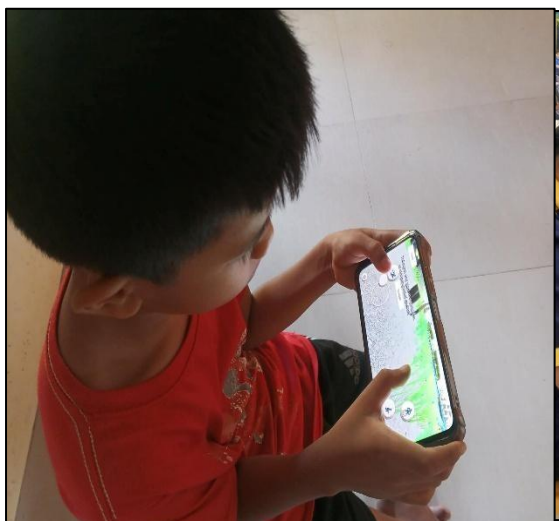
USABILITY	1	2	3	4	5
General Usability Questions:					✓
1. How easy was it to understand the game's controls?					✓
2. Were the instructions clear before starting the game?					✓
3. How clear were the game's objectives (e.g., interacting with characters, finding artifacts and complete missions)?				✓	
Gameplay Experience Questions:					
4. Did the Bolinao quests (Bolinao Skull, First Mass, quiz, cutscene) clearly explain the history?				✓	
5. Did the Bani quests (Origin of the name Bani, crops/farming, quiz, cutscene) clearly explain the history?				✓	
6. Were the controls and interactions (movement, dialogue, quests) easy to use and understand?					✓
Interface and Visual and Design:					
7. Did the game design (graphics, environment, and layout) help you focus on the historical content?					✓
Performance and Technical Question:					
8. Did the game run smoothly without lag or crashes?				✓	
Overall Usability and Satisfaction:					
9. Did you feel engaged and interested while playing the game?					✓
10. Overall, was your experience playing the game is it enjoyable and easy to use as a digital tool to learn history?					✓

Year level and set: 4th year

John Van Geronimo

Signature over the Printed Name

APPENDIX F SURVEY DOCUMENTATION





APPENDIX G SOURCE CODE



https://drive.google.com/drive/folders/1s_P1gS0n0MHhFx7ff0pnYk-9JQ3vyDkc



How to install & operate the game:

- Scan the provided QR code or access the link to download the game file.
- Once redirected to the download page, download the game application file (APK).
- After the download is complete, open the APK file on the mobile device.
- If prompted, allow the installation from unknown sources to proceed with the installation.
- Tap Install and wait for the installation process to finish.
- Once installed, open the application on the mobile device to start the game.



CURRICULUM VITAE



NEIL MATTHEW R. AQUINO

BSIT Student

Contact

0926 968 7632
 neil.aquino.gwc@gmail.com
 Poblacion East Sual, Pangasinan

About Me

"An IT student that loves to progress his knowledge."

Skills

Beginner level

- Web development
- App development
- Graphic Design

Education

Agbayani Street, Poblacion East Sual, Pangasinan

- Sual, Central, School 2009 – 2016
 Sual, National, High School 2016 - 2020
 Sual, Senior, High School 2020 - 2022
General Academic Strand
- San Jose Drive, Alaminos City
 Golden West College's 2022 - present
Bachelor of Science in Information Technology

Language

- Tagalog
- English

Personal Details

- Full Name: Neil Matthew Reandino Aquino
 Birthdate: June 7, 2003
 Age: 22
- Civil Status: Single
 Sex: Male
- Height: 5'5
 Religion: Roman Catholic
- Nationality: Filipino

Character References

Mike S. Enriquez
 Electrical Engineer

 Cp #: 0915 905 092



ALEXIS MAE A. DELA CRUZ

BSIT Student

Contact

- ☎ 0967 177 3666
- ✉ abinaalexis96@gmail.com
- 📍 015 Purok 4, Sampaloc, Bolinao, Pangasinan

About Me

"A motivated college student with strong interest in learning new skills and gain practical experience. I am responsible, and eager to contribute to a professional environment while continuing to grow academically and personally."

Skills

- Good Listener
- Willing to be trained
- Hand Drawing
- Basic Graphic Design and Visual in Canva

Language

- Tagalog
- English
- Ilocano

Education

- Sampaloc, Bolinao, Pangasinan
 - Sampaloc, Elementary, School 2010 – 2013
 - Sampaloc, National, High School 2013 - 2017
 - Sampaloc, Senior, High School 2018 - 2020
 - Information Communication Technology
- San Jose Drive, Alaminos City
 - Golden West College's 2022 - present
 - Bachelor of Science in Information Technology

Personal Details

- Full Name: Alexis Mae A. Dela Cruz
- Birthdate: May 9, 2002
- Age: 23
- Civil Status: Single
- Sex: Female
- Height: 5'3
- Religion: Roman Catholic
- Nationality: Filipino

Character References

Edna D. Degracia
Accounting Manager in
Xworks Sales & Services, Inc.
Baguio City

Cp #: 0935 122 8423

FB account: Edna Dela Cruz Degracia



Contact

☎ 0950 840 2203
 @ kazandraturgade@gmail.com
 📍 01 Purok 1, Cabuyao, Bolinao, Pangasinan

Skills

- Good Listener
- Calligraphy
- Basic Video Editing
- Photography

Language

- Tagalog
- English
- Ilocano

Character References

Elma Kathleen Sioson
 Senior Treasury Supervisor in
 Masaito Development Corporation
 Makati City

Cp #: 0933 011 9116

About Me

"Looking forward to working in a role where I can apply my knowledge, adapt to new challenges, and contribute to organizational success while preparing for future career opportunities".

Education

Cabuyao, Bolinao, Pangasinan

• Cabuyao, Elementary School 2009 – 2016

Luna, Bolinao, Pangasinan

Luna, National High School 2016 - 2020

Luna, Senior High School 2020 - 2022

Technical Vocational Livelihood

San Jose Drive, Alaminos City

Golden West College's 2022 - present

Bachelor of Science in Information Technology

Personal Details

- Full Name: Kazandra Roxas Tugade
- Birthdate: April 16, 2004
- Age: 21
- Civil Status: Single
- Sex: Female
- Height: 5'4
- Religion: Roman Catholic
- Nationality: Filipino



Julius S. Ferrer

BSIT Student

Contact

☎ 0981 729 3459
 @ juliusferrer108@gmail.com
 📍 San Bruno, Bangan Oda, Agno
 Pangasinan

Skills

- Network Setup

Language

- Tagalog
- Ilocano

Character References

Mhelody S. Nuevo-Nurse- Agno
 Phone: 0928 071 7167

About Me

"As a motivated student, I aim to join a company where I can apply my passion for learning, develop new skills, and contribute positively to team goals".

Education

- Mabini, Street, Poblacion West, Pangasinan
- Agno Central, School 2009-2016
 Agno, National, High School 2016 - 2020
 Agno, Senior, High School 2020 - 2022
 Technical-Vocational-Livelihood
 - San Jose Drive, Alaminos City
 Golden West College's 2022 - present
 Bachelor of Science in Information Technology

Personal Details

- Full Name: Julius Sagun Ferrer
 Birthdate: July 9, 2004
 Age: 21
- Civil Status: Single
 Sex: Male
 Height: 5'3
 Religion: Roman Catholic
- Nationality: Filipino

